



Tennessee TECH

Default Question Block

Pre-Activity Quiz Prior to Flagmaker Activity

Multiple Choice: What is the primary limitation of using only one processor to complete a task?

- It uses too much memory, making the process slower
 - It can only access local files, making the process slower
 - It must complete each step one at a time, making the process slower
 - It causes data corruption
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Multiple Choice: What is a potential drawback of using multiple processors?

- Multiple processors can cause overheating requiring expensive cooling techniques
 - Multiple processors can cause the computer to run out of RAM
 - The processors may finish at different times, causing some to sit idle with no work
 - The processors are able to complete the task faster
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True or False: Speedup is defined as the ratio of the time taken to solve a problem on a single processor to the time taken on a parallel system.

- True
 - False
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Multiple Choice: What is contention in parallel computing?

- The process of dividing (breaking down) a task into smaller sub-tasks.
- When multiple processors are waiting on (competing for) shared resources.

The increase in computational speed by adding more processors.

The ability of a system to handle a growing amount of work.

True or False: Scalability refers to the ability of a parallel system to increase its performance proportionally with the addition of more processors.

True

False

Multiple Choice: What is the term for ensuring that each processor has roughly the same amount of work to prevent idle time?

Multithreading

Synchronization

Load balancing

Memory sharing

Multiple Choice: What is a race condition in the context of parallel computing?

Where one processor finishes the task before others start

Where multiple processors try to access or modify shared data simultaneously

A scheduling technique that improves processor speed and synchronization

A timeout that resets processor tasks

Multiple Choice: What is pipelining in the context of parallel computing?

The process of executing multiple tasks simultaneously by multiple processors.

The strategy of reducing contention among processors so that multiple processors can't access the same shared resource at the same time.

The method of coordinating multiple processors to ensure they work together correctly when accessing shared resources.

The technique of dividing a task into sub-tasks and each subtask is handled by a different processor in a sequence like an assembly line.

Enter your **TTU Username** (example: jdoe42)